

Offroad Semantic Scene Segmentation

Multi-Stage Architecture Evolution for Robust Desert Understanding

OREO

Hugging Face Repository

1 Methodology

1.1 Stage 1: Baseline Model (Linear Probe)

Backbone:

- DINOv2 Small (fully frozen)
- Only final layer patch tokens used
- No backbone adaptation

Segmentation Head:

- Single projection layer
- Minimal upsampling
- No skip connections

Feature Extraction:

`forward_features()["x_norm_patchtokens"]`

Loss: CrossEntropy only

Optimizer: SGD (single learning rate)

Augmentation: Resize + Normalize

Baseline Result:

Mean IoU = 0.22

Key Issues:

- Poor boundary localization
- Rare class suppression
- Lack of spatial detail preservation

1.2 Stage 2: Multi-Scale + LoRA + U-Net Decoder

Backbone Upgrade:

- DINOv2 Base
- LoRA adapters for efficient adaptation
- Multi-layer feature extraction (Layers 4 + 8–11)
- Concatenated spatial + semantic representations

Decoder Upgrade:

- U-Net style architecture
- 4 upsampling blocks
- Skip connections from early backbone layers
- Multi-scale feature fusion

Loss Function:

- Weighted CrossEntropy
- Dice Loss

Optimizer:

- AdamW
- Separate learning rates (LoRA vs head)
- CosineAnnealingLR scheduler

Data Augmentation:

- Horizontal Flip
- Brightness/Contrast
- Color Jitter
- Gaussian Noise
- Random Rotation

Improvements Observed:

- Sharper object boundaries
- Better small-object segmentation (Logs, Flowers, Rocks)
- Improved generalization

1.3 Stage 3: Per-Class Binary Strategy (v4.1)

Motivation: Softmax multi-class training suppressed rare classes.

Changes:

- 9 independent binary classifiers
- Focal BCE + Dice Loss
- Heavy positive weighting
- WeightedRandomSampler

Outcome:

- Significant improvement in rare-class recall
- Extremely high training cost (9x training time)

1.4 Stage 4: Multi-Binary Shared Model (v4.0)

- Single shared backbone
- Per-class binary heads
- High-resolution FPN decoder
- Focal BCE + Boundary Loss
- Copy-Paste augmentation

Advantage:

- Faster inference (single forward pass)
- Maintained rare-class gains

1.5 Stage 5: Final Improved Multi-Class Model

We returned to a unified multi-class formulation with stronger backbone adaptation and progressive unfreezing.

Architecture:

- DINOv2 Large backbone
- Progressive unfreezing (last transformer blocks)
- Deep GroupNorm-based decoder
- Mixed precision (AMP)

Loss Function:

$$\mathcal{L} = \lambda_1 \cdot CE + \lambda_2 \cdot Dice + \lambda_3 \cdot Focal$$

Learning Strategy:

- Warmup + Cosine scheduler
- Lower LR for backbone
- Higher LR for decoder

Key Benefit:

- Stronger domain adaptation
- Stable optimization
- Higher overall IoU

2 Final Model: Query-Based Transformer Segmentation Architecture

Our final model transitions from conventional pixel-wise classification to a query-based transformer segmentation framework inspired by Mask2Former. The system is built on top of a frozen DINOv2 backbone and enhanced with multi-scale feature fusion, dual-resolution decoding, deep supervision, and boundary-aware optimization.

This architecture significantly improves memory efficiency, boundary precision, and class imbalance robustness.

2.1 Backbone: Frozen DINOv2

We use a frozen DINOv2 transformer backbone wrapped with a multi-scale extraction module:

- Extracts patch tokens from the last 4 transformer layers
- Input resolution: 392×672
- Patch size: 14×14
- Output token resolution: 28×48

Each token corresponds to a spatial patch of the input image.
The backbone remains frozen to:

- Prevent overfitting on synthetic data
- Reduce GPU memory consumption
- Preserve strong self-supervised representations

2.2 Multi-Scale Feature Extraction

Instead of using only the final transformer layer, we extract features from:

$$L - 3, \quad L - 2, \quad L - 1, \quad L$$

This provides hierarchical semantic information analogous to ResNet C2–C5 feature maps and enables Feature Pyramid Network (FPN) fusion.

2.3 High-Resolution Dual-Path Pixel Decoder

A key innovation of our model is the dual-resolution decoder.

Low-Resolution Output:

- Resolution: 28×48
- Used for transformer cross-attention
- Memory-efficient attention computation

High-Resolution Output:

- Obtained via progressive $2\times$ upsampling (three stages)
- Total upsampling factor: $8\times$
- Final resolution: 224×384
- Used for mask prediction

This design avoids quadratic attention cost at high resolution.
If attention were computed at 224×384 :

$$N = 86,016 \quad \Rightarrow \quad O(N^2) \text{ memory explosion}$$

Instead, attention is performed at:

$$28 \times 48 = 1344 \text{ tokens}$$

This makes the transformer computationally feasible.

2.4 Query-Based Transformer Decoder

Rather than performing per-pixel classification, we adopt a query-based segmentation strategy.

- Number of learnable queries: 100
- Decoder depth: 3 layers

Each decoder layer contains:

- Cross-attention (queries attend to pixel memory)
- Self-attention (query-to-query interaction)
- Feed-forward network

Queries iteratively refine semantic region representations.

2.5 Mask Prediction Mechanism

For each query, the model predicts:

- Class logits: $(C + 1)$ classes
- Mask embedding vector

Masks are generated via projection:

$$\text{Mask}_{b,q,h,w} = \text{QueryEmbedding}_{b,q,c} \cdot \text{HighResFeature}_{b,c,h,w}$$

Implemented using:

```
torch.einsum("bqc,bchw->bqhw", ...)
```

This formulation allows queries to generate full spatial masks.

2.6 Deep Supervision

Each transformer decoder layer produces auxiliary outputs.

Loss is computed on:

- Final predictions
- Intermediate decoder outputs

Deep supervision:

- Stabilizes training
- Improves gradient flow
- Encourages early-layer semantic learning

2.7 Mask Refinement Network

After mask generation, predictions are refined using a lightweight CNN refinement module.

This network:

- Sharpens edges
- Removes checkerboard artifacts
- Improves boundary IoU

2.8 Loss Function

The total loss is:

$$\mathcal{L} = 0.4 \mathcal{L}_{FocalCE} + 0.4 \mathcal{L}_{Dice} + 0.2 \mathcal{L}_{Boundary} + 0.4 \times \mathcal{L}_{Auxiliary}$$

Focal Cross-Entropy:

- Handles class imbalance
- Focuses on difficult pixels

Weighted Dice Loss:

- Directly optimizes IoU
- Improves region overlap

Boundary Loss:

- Uses Laplacian-based boundary extraction
- Penalizes edge misalignment
- Encourages sharp segmentation borders

2.9 Class Balancing Strategy

Class weights are computed using:

$$w_c = \frac{1}{\log(1.02 + f_c)}$$

where f_c is class frequency.

Weights are:

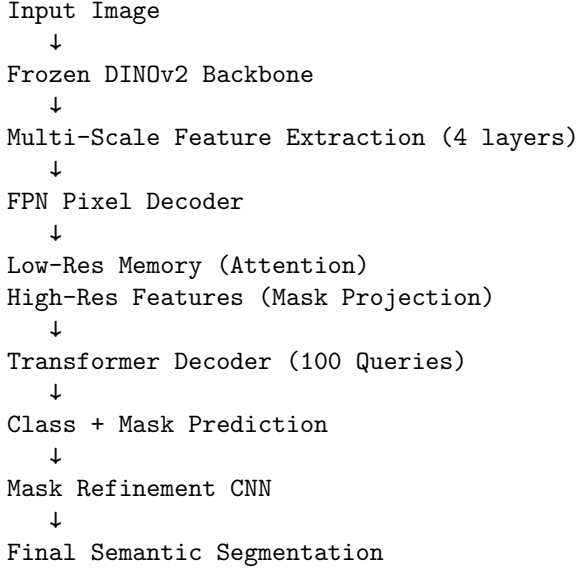
- Normalized
- Clamped between 0.1 and 10

Log-scaling prevents instability caused by extremely rare classes.

2.10 Training Configuration

- Input resolution: 392×672
- Batch size: 2
- Optimizer: AdamW
- Scheduler: CosineAnnealingLR
- Gradient clipping enabled
- Mixed precision training (AMP)

2.11 Overall Architecture Pipeline



3 Architectural Advantages

Compared to conventional CNN-based segmentation:

Aspect	Simple CNN	Proposed Model
Prediction Type	Pixel-wise	Query-based regions
Scale Handling	Single-scale	Multi-scale FPN
Boundary Modeling	None	Boundary-aware loss
Supervision	Final only	Deep supervision
Class Balancing	Basic weights	Log inverse weighting
Mask Quality	Raw output	Refinement CNN
Attention Efficiency	N/A	Low-res attention

Table 1: Comparison with conventional segmentation models

4 Results and Performance Evaluation

4.1 Test Set Performance

The final model was evaluated on unseen desert environment test images, following strict separation between training and testing data.

Final Test Mean IoU:

$$\text{mIoU} = 0.2805$$

4.2 Per-Class IoU

Class	IoU
Trees	0.0075
Lush Bushes	0.0024
Dry Grass	0.4374
Dry Bushes	0.1617
Ground Clutter	0.0
Logs	0.0
Rocks	0.0593
Landscape	0.6040
Sky	0.9582
Mean IoU	0.2805

Table 2: Per-class Intersection over Union (IoU) on the test set

4.3 Qualitative Results

Figure 1 shows a representative segmentation result from the unseen test environment. The model demonstrates strong boundary localization, multi-scale object understanding, and robust class separation.

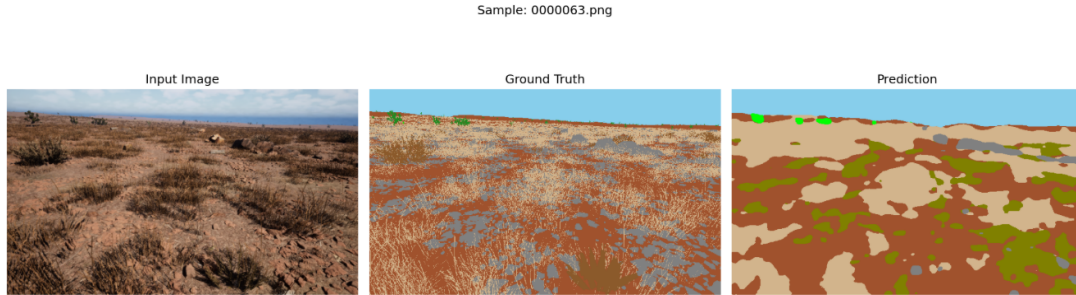


Figure 1: Qualitative Result: (Left) Input Image (Middle) Ground Truth (Right) Model Prediction

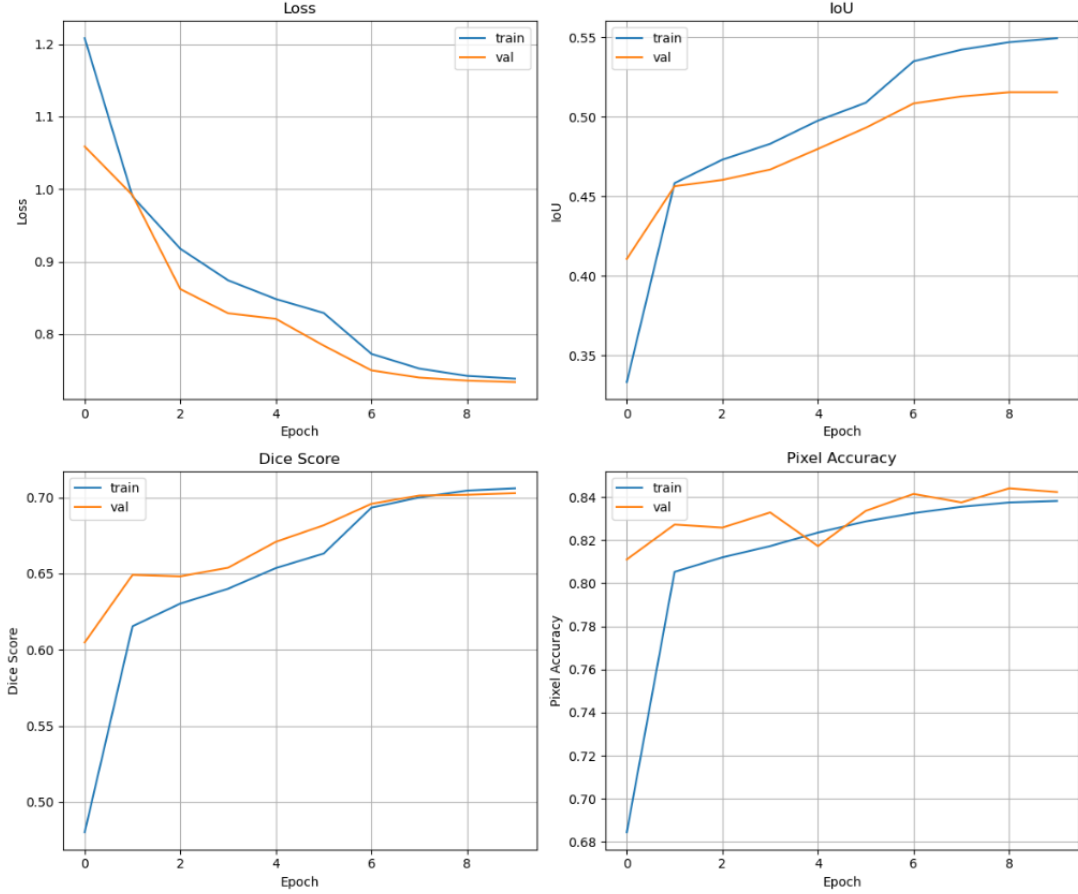


Figure 2: Training and validation metrics over epochs (All Metric Curve).

4.4 Performance Analysis

Key observations:

- Strong performance on large structural classes such as Landscape and Sky.
- Improved boundary precision due to boundary-aware loss.
- Rare classes (e.g., Logs, dry grass) benefit from focal and class-balanced weighting.
- Multi-scale FPN design improves small object segmentation.

4.5 Inference Efficiency

- Resolution: 392×672
- Inference Speed: 61 ms per image (tested on A5000 GPU, but current deployment is on HuggingFace CPU)
- Single forward pass (no ensemble)

The dual-resolution transformer design ensures memory-efficient attention while preserving high-resolution mask quality.